

Anshika Shukla

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EDUCATION

B.Tech, Electronics & Communication Engineering — Banasthali Vidyapith | Aug 2023 – May 2027

4th Year | CGPA: 7.38/10 | Specialization: Edge AI & Embedded Systems

KEY ACHIEVEMENTS

- 2nd Runner-Up, HACKVERSE — National Hackathon (100+ teams); Edge AI emergency platform
- Top 10, HACKCELESTIA — National Hackathon; Satellite-relay AI system
- Selected, Sanchar Saathi Initiative — Govt. of India, Dept. of Telecommunications
- GSSoC 2026 Open Source Contributor — active contributor across multiple repositories
- Poster Presenter, WACIIS 2025 International Conference
- Preprint: "Systematic Review on Brain-Computer Interfaces" — SSRN ID: 6763558, 2025

PROJECTS

PitWall — F1 Race Intelligence Platform | XGBoost | FastAPI | Next.js | Three.js | FastF1 | Python | Jun 2026

- Designed end-to-end ML pipeline ingesting FastF1 telemetry across 5 seasons (2,449 rows); XGBoost model achieved cross-val MAE 3.61 positions predicting race outcomes for all 20 drivers
- Engineered DriverStrength composite signal (DriverForm 40% + TrackHistory 30% + ChampionshipPoints 30%); bias correction reduced front-row underestimation; QualiPosition dominance cut from 39.8% to 12.8%
- Built FastAPI inference server + Next.js frontend with Three.js 3D visualization and live OpenMeteo weather feed; Monaco Ex-DNF MAE 2.8 — top-3 predicted within 1 position each

SHL Assessment Recommender | Python | NLP | FastAPI | Semantic Search | Sentence Transformers | May 2026

- Built end-to-end semantic recommendation system: sentence transformer embeddings, cosine similarity ranking, FastAPI deployment — returns ranked SHL assessments with relevance scores per job description query
- Structured as modular pipeline (embed → rank → serve) enabling swap of embedding model without API changes; tested across varied job description lengths and formats

Flight-Sense-AI — Aviation Anomaly Detection | Python | Scikit-learn | Sensor Telemetry | FastAPI | Time-Series ML | Live Flight Data | OpenSky Network ADS-B REST API | May 2026

- Developed real-time aviation anomaly detection system processing multi-sensor telemetry streams; ML classifier identifies flight parameter deviations indicative of system faults
- Pipeline covers sensor ingestion, feature extraction from time-series data, and anomaly scoring — relevant to satellite telemetry health monitoring use cases

SMART-GUARD — IoT Safety Helmet | Arduino | GSM | GPS | IoT | MPU6050 | MQ3 | Apr 2026

- Engineered embedded 4-sensor fusion system (MPU6050 accelerometer, IR proximity, MQ3 alcohol, UV index) on Arduino with GSM/GPS delivering emergency location SMS within 2 seconds of impact detection
- Recognized by Govt. of India Sanchar Saathi Initiative (Dept. of Telecommunications) for embedded safety communication design; national-level selection

SatLink — Satellite-Relay Edge AI Emergency Platform | Edge AI | Python | Raspberry Pi | Satellite Relay | IoT | Jan 2026

- Architected offline-first emergency AI platform routing inference through satellite relay for connectivity-denied regions; designed for zero cloud dependency in disaster-response scenarios
- Achieved low-latency relay pipeline in hackathon demo environment — Top 10 nationally at HACKCELESTIA; architecture relevant to satellite-ground edge inference deployment challenges

SignEase — Real-Time Sign Language Translator | TensorFlow Lite | Raspberry Pi | OpenCV | MediaPipe | CNN | Sep 2025

- Compressed CNN by 81% via post-training quantization and pruning (45MB → 8.5MB); achieved sub-100ms inference on Raspberry Pi 4 without GPU — directly applicable to satellite payload compute constraints
- TFLite deployment with MediaPipe landmark pipeline; sustained 24 FPS at 91% accuracy on Class-4 constrained hardware — demonstrates model optimization for edge deployment

SafeGaze — AI Content Moderation System | TensorFlow | OpenCV | FastAPI | Python | REST API | Mar 2025

- Trained content moderation classifier to 95% test accuracy; batch inference pipeline achieves 847 images/sec on CPU benchmark — FastAPI deployment wraps model as production-ready REST microservice
- Modular architecture: classification backbone decoupled from serving layer; supports model hot-swap without API downtime

TECHNICAL SKILLS

Programming Languages: Python, C, C++, JavaScript, TypeScript, SQL

ML & AI Frameworks: TensorFlow, PyTorch, XGBoost, scikit-learn, TensorFlow Lite, OpenCV, MediaPipe

Edge & Embedded: Raspberry Pi, Arduino, ESP32, TFLite deployment, IoT sensor fusion, MQTT

Satellite & Comms: Satellite relay systems, GSM/GPS integration, offline-first architecture

Backend & APIs: FastAPI, REST APIs, Uvicorn, joblib, pandas, NumPy, Git

Data & ML Engineering: Feature engineering, model compression, quantization, pruning, FastF1, data pipelines

Tools & Platforms: VS Code, Google Colab, Jupyter, MATLAB, GitHub, Linux, Overleaf

CS Fundamentals: Data Structures & Algorithms, OOP, Computer Networks, Operating Systems, DBMS

Languages (Spoken): English (Proficient), Hindi (Native), French (B1)

CERTIFICATIONS & TRAINING

- Building Data & AI Solutions on Google Cloud — 365 Data Science (100%, Mar 2026)
- Generative AI — Microsoft & LinkedIn Learning
- NASA ARSET Remote Sensing Training Program
- MATLAB Onramp — MathWorks
- Drone Pilot Certification — Banasthali Vidyapith
- French B1 Diploma